Chapter 11: Simple Linear Regression and Correlation

ENGR 0021

Learning Goals

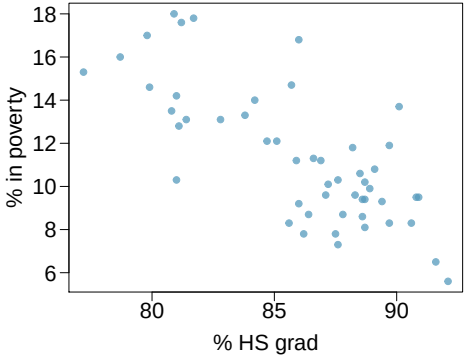
- Model relationships between variables
- Interpret regression coefficients
- Assess model fit and assumptions
- Perform inference for the slope

Modeling numerical variables

We will learn to quantify the relationship between two numerical variables, as well as modeling numerical response variables using a numerical or categorical explanatory variable.

Poverty vs. HS graduate rate

The **scatterplot** below shows the relationship between HS graduate rate in all 50 US states and DC and the % of residents who live below the poverty line (income below $23,050 for a family of 4 in 2012).



Response variable?

Explanatory variable?

Relationship?

Linear Model

The linear model for predicting poverty from high school graduation rate in the US is

$$\hat{poverty} = 64.78 - 0.62 * HS_{grad}$$

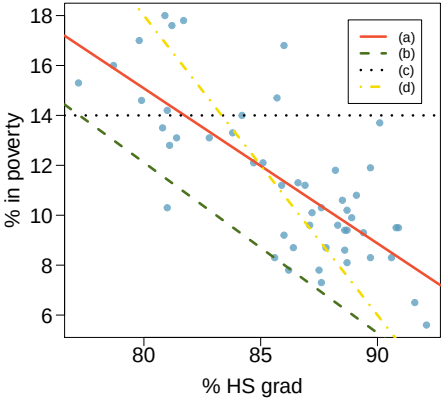
The “hat” is used to signify that this is an estimate.

Linear Model

The high school graduate rate in Georgia is 85.1%. What poverty level does the model predict for this state?

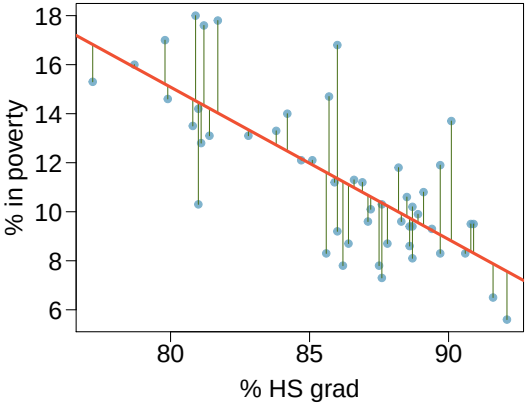
Eyeballing the line

Which of the following appears to be the line that best fits the linear relationship between % in poverty and % HS grad? Choose one.



Residuals

Residuals are the leftovers from the model fit: Data = Fit + Residual

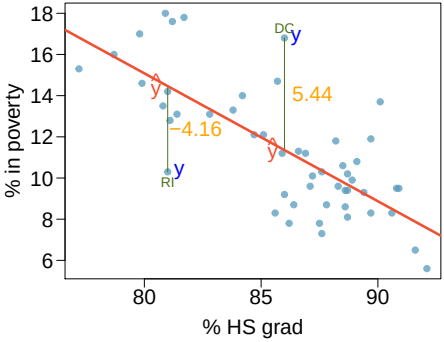


Residuals (cont.)

Residual

Residual is the difference between the observed (y_i) and predicted $\hat{y}_i$.

$$e_i = y_i - \hat{y}_i$$



- % living in poverty in DC is 5.44% more than predicted.
- % living in poverty in RI is 4.16% less than predicted.

Quantifying the relationship

- **Correlation** describes the strength of the **linear** association between two variables.
- The sample correlation coefficient is

$$r = \frac{S_{XY}}{\sqrt{S_{XX}S_{YY}}}.$$

- It takes values between -1 (perfect negative) and +1 (perfect positive).
- A value of 0 indicates no linear association.

Correlation Coefficient

- ρ: population correlation coefficient
- r: sample correlation coefficient
- r is a point estimate of ρ

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

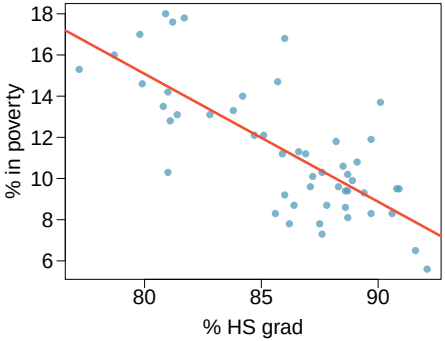
Properties of the Correlation Coefficient

- r does not depend on which variable is labeled x or y.
- r is unitless.
- $$-1 \leq r \leq 1$$
- $r = 1$ iff all points lie exactly on a straight line with positive slope.
- $r = -1$ iff all points lie exactly on a straight line with negative slope.

Guessing the correlation

Which of the following is the best guess for the correlation between % in poverty and % HS grad?

(a) 0.6
(b) -0.75
(c) -0.1
(d) 0.02
(e) -1.5



Guessing the correlation

Which of the following is the best guess for the correlation between % in poverty and % female householder, no husband present?

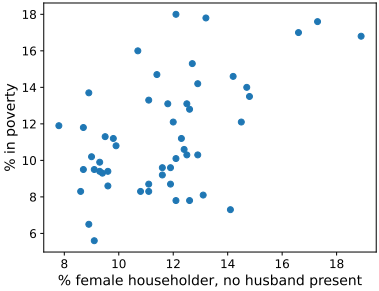
(a) 0.1

(b) -0.6

(c) -0.4

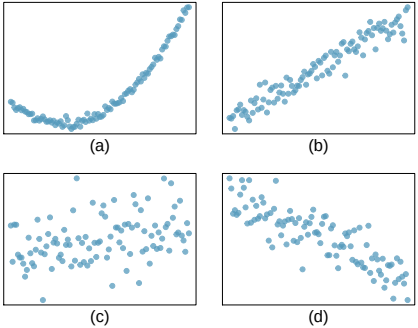
(d) 0.9

(e) 0.5



Assessing the correlation

Which of the following is has the strongest correlation, i.e. correlation coefficient closest to +1 or -1?



An objective measure for finding the best line

- We want a line that has small residuals:
  1. Option 1: Minimize the sum of magnitudes (absolute values) of residuals

$$|e_1| + |e_2| + \cdots + |e_n|$$

  2. Option 2: Minimize the sum of squared residuals – **least squares**
     SSE =

$$e_1^2 + e_2^2 + \cdots + e_n^2$$

- Why least squares?
  1. Most commonly used
  2. Easier to compute by hand and using software
  3. In many applications, a residual twice as large as another is usually more than twice as bad

The least squares line

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

$$Y \sim N(\beta_0 + \beta_1 x, \sigma^2)$$

- $\hat{y}$: Predicted value of the response variable, y
- β_0: Intercept, parameter
  - b_0: Intercept, point estimate
- β_1: Slope, parameter
  - b_1: Slope, point estimate
- x: Explanatory or predictor variable

The Simple Linear Regression Model

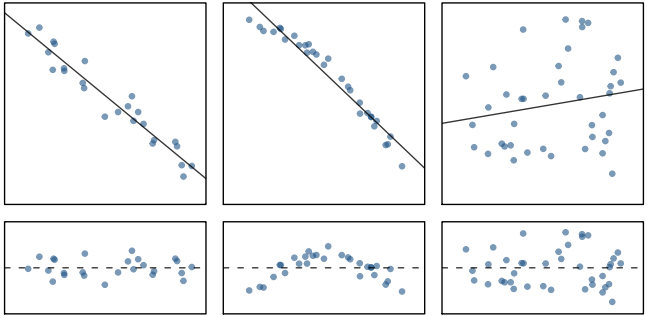
$$\hat{y} = b_0 + b_1 x$$

Conditions for the least squares line

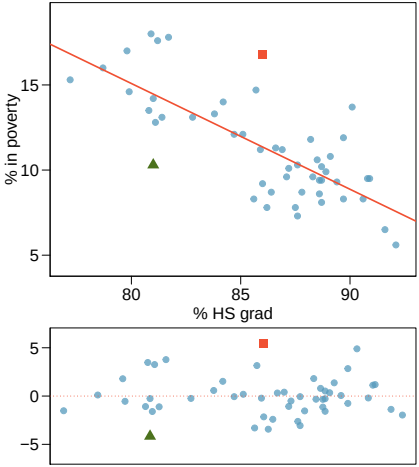
1. Linearity
2. Nearly normal residuals
3. Constant variability

Conditions: (1) Linearity

- The relationship between the explanatory and the response variable should be linear.
- Methods for fitting a model to non-linear relationships exist, but are beyond the scope of this class.
- Check using a scatterplot of the data, or a **residuals plot**.



Anatomy of a residuals plot



▲ **RI:**

$$\%\ HS\ grad = 81 \qquad \%\ in\ poverty = 10.3$$

$$\widehat{\%\ in\ poverty} = 64.68 - 0.62 * 81 = 14.46$$

$$e = \%\ in\ poverty - \widehat{\%\ in\ poverty}$$

$$= 10.3 - 14.46 = -4.16$$

■ **DC:**

$$\%\ HS\ grad = 86 \qquad \%\ in\ poverty = 16.8$$

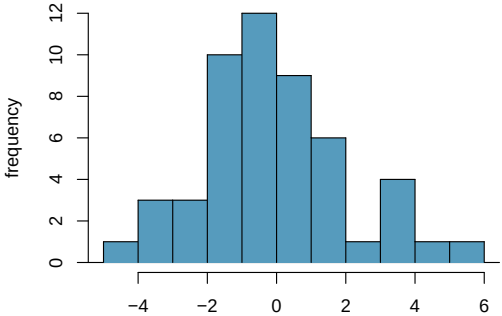
$$\widehat{\%\ in\ poverty} = 64.68 - 0.62 * 86 = 11.36$$

$$e = \%\ in\ poverty - \widehat{\%\ in\ poverty}$$

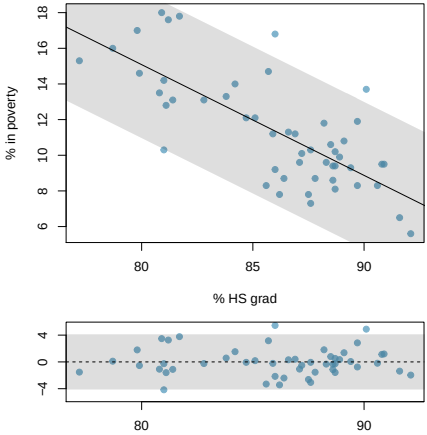
$$= 16.8 - 11.36 = 5.44$$

Conditions: (2) Nearly normal residuals

- The residuals should be nearly normal.
- This condition may not be satisfied when there are unusual observations that don't follow the trend of the rest of the data.
- Check using a histogram.



Conditions: (3) Constant variability

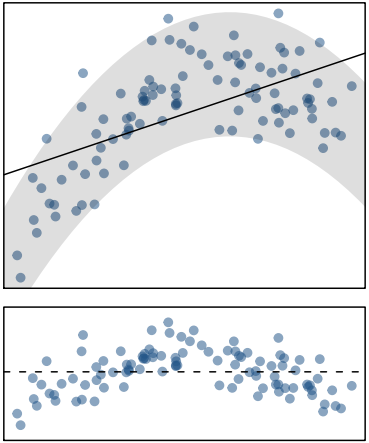


- The variability of points around the least squares line should be roughly constant.
- This implies that the variability of residuals around the 0 line should be roughly constant as well.
- Also called **homoscedasticity**.
- Check using a residuals plot.

Checking conditions

What condition is this linear model obviously violating?

(a) Constant variability
(b) Linear relationship
(c) Normal residuals
(d) No extreme outliers



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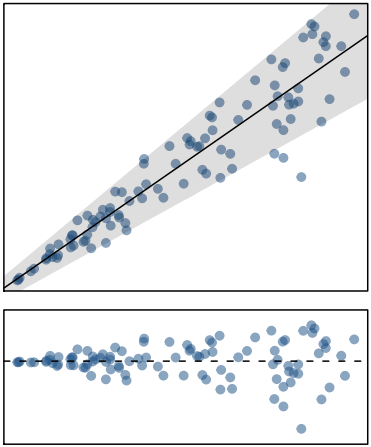
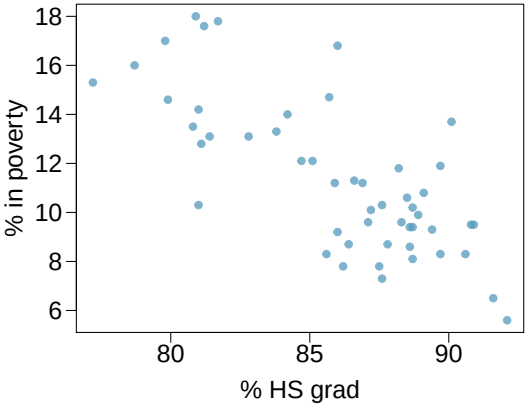


Given...



	% HS grad (x)	% in poverty (y)
mean	$\bar{x} = 86.01$	$\bar{y} = 11.35$
sd	$s_x = 3.73$	$s_y = 3.1$
	correlation	$r = -0.75$

Useful Sums

Define

$$S_{xx} = \sum (x_i - \bar{x})^2, \qquad S_{yy} = \sum (y_i - \bar{y})^2, \qquad S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

Sample standard deviations:

$$s_x = \sqrt{\frac{S_{xx}}{n-1}}, \qquad s_y = \sqrt{\frac{S_{yy}}{n-1}}$$

These summarize variability and association in the data.

Slope of the Regression Line

Equivalent formulas:

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{s_y}{s_x} r$$

In context:

$$b_1 = \frac{3.1}{3.73} \times (-0.75) = -0.62$$

Interpretation:

For each additional % point in HS graduation rate, the % living in poverty decreases by about 0.62 percentage points on average. With each increase of one s_x there is an increase of only $r\ s_y$, on the average.

Mean Squared Error

- SSE has $n - 2$ degrees of freedom.
- Define
$$s^2 = MSE = \frac{SSE}{n-2}$$
- Equivalently,
$$MSE = \frac{S_{yy} - b_1 S_{xy}}{n-2}$$
- MSE is the estimator of the error variance.

Intercept of the Regression Line

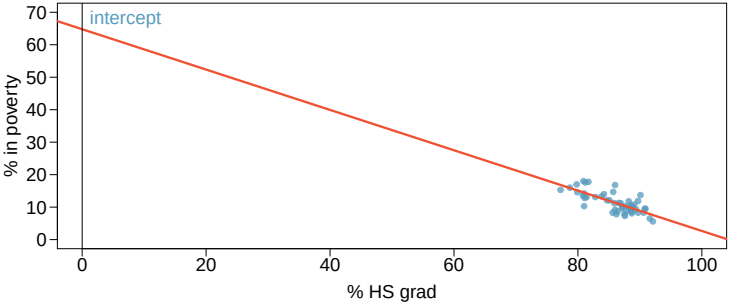
$$b_0 = \bar{y} - b_1 \bar{x}$$

Interpretation:
Predicted value of y when $x = 0$.

Intercept

The intercept is where the regression line intersects the y-axis. The calculation of the intercept uses the fact the a regression line always passes through $(\bar{x}, \bar{y})$.

$$b_0 = \bar{y} - b_1 \bar{x}$$



$$\begin{aligned} b_0 &= 11.35 - (-0.62) \times 86.01 \\ &= 64.68 \end{aligned}$$

Intercept Interpretation

Which of the following is the correct interpretation of the intercept?

(a) For each % point increase in HS graduate rate, % living in poverty is expected to increase on average by 64.68%.

(b) For each % point decrease in HS graduate rate, % living in poverty is expected to increase on average by 64.68%.

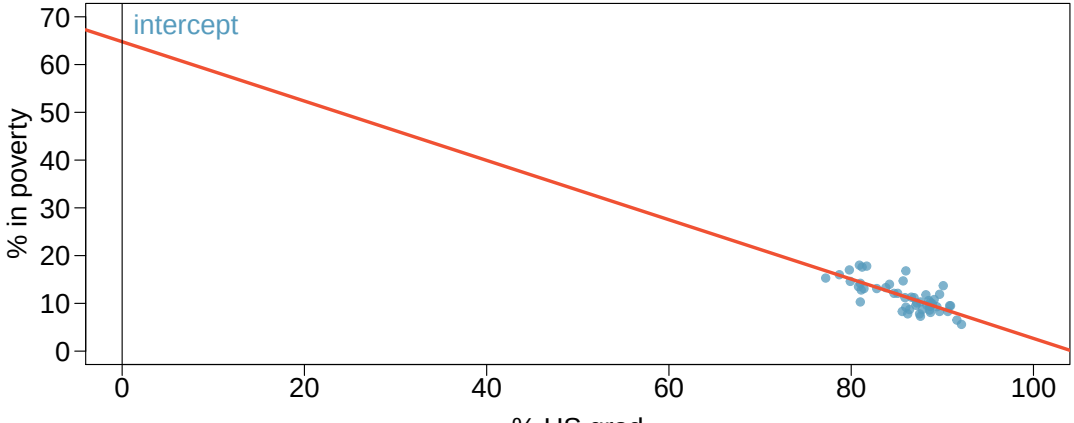
(c) Having no HS graduates leads to 64.68% of residents living below the poverty line.

(d) States with no HS graduates are expected on average to have 64.68% of residents living below the poverty line.

(e) In states with no HS graduates % living in poverty is expected to increase on average by 64.68%.

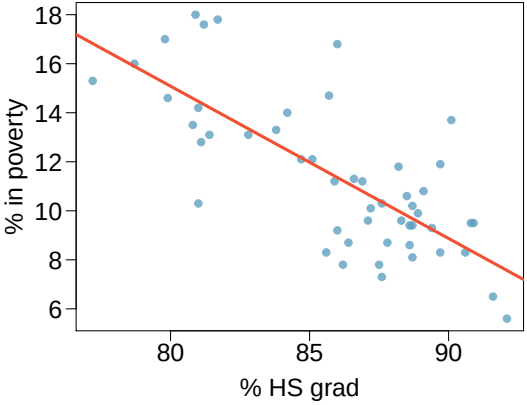
More on the intercept

Since there are no states in the dataset with no HS graduates, the intercept is of no interest, not very useful, and also not reliable since the predicted value of the intercept is so far from the bulk of the data.



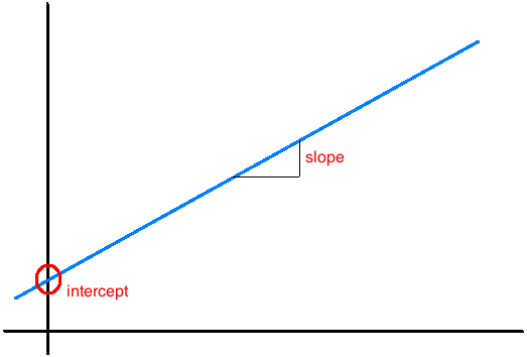
Regression line

$$\widehat{\%\ in\ poverty} = 64.68 - 0.62\ \%\ HS\ grad$$



Interpretation of slope and intercept

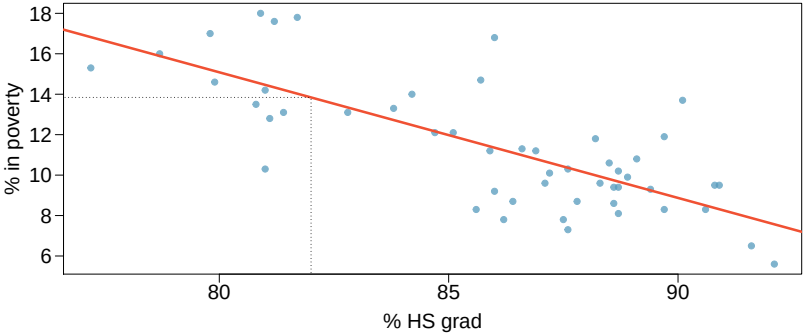
- **Intercept:** When $x = 0$, y is expected to equal the intercept.
- **Slope:** For each unit in x, y is expected to increase / decrease on average by the slope.



Note: These statements are not causal, unless the study is a randomized controlled experiment.

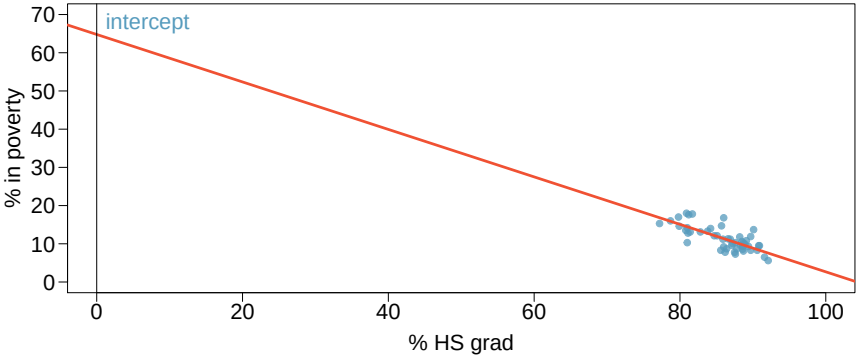
Prediction

- Using the linear model to predict the value of the response variable for a given value of the explanatory variable is called **prediction**, simply by plugging in the value of x in the linear model equation.
- There will be some uncertainty associated with the predicted value.

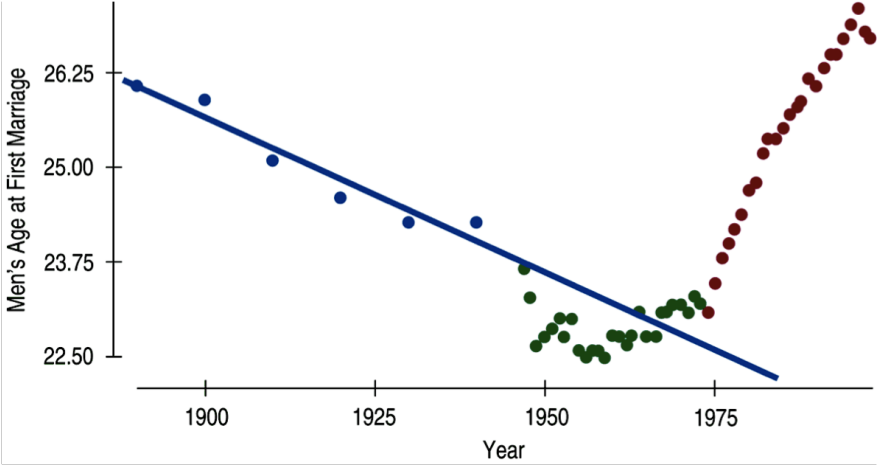


Extrapolation

- Applying a model estimate to values outside of the realm of the original data is called **extrapolation**.
- Sometimes the intercept might be an extrapolation.



Examples of extrapolation



Examples of extrapolation

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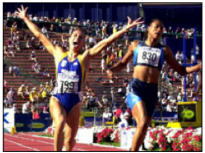
Women 'may outsprint men by 2156'

Women sprinters may be outrunning men in the 2156 Olympics if they continue to close the gap at the rate they are doing, according to scientists.



Women are set to become the dominant sprinters

An Oxford University study found that women are running faster than they have ever done over 100m.

At their current rate of improvement, they should overtake men within 150 years, said Dr Andrew Tatem.

The study, comparing winning times for the Olympic 100m since 1900, is published in the journal Nature.

However, former British Olympic sprinter Derek Redmond told the BBC: "I find it difficult to believe.

"I can see the gap closing between men and women but I can't necessarily see it being overtaken because mens' times are also going to improve."

Examples of extrapolation

Momentous sprint at the 2156 Olympics?

Women sprinters are closing the gap on men and may one day overtake them.

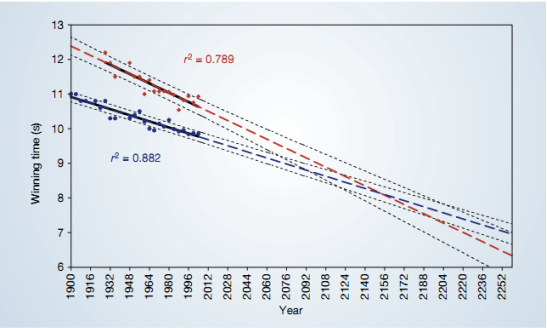


Figure 1 The winning Olympic 100-metre sprint times for men (blue points) and women (red points), with superimposed best-fit linear regression lines (solid black lines) and coefficients of determination. The regression lines are extrapolated (broken blue and red lines for men and women, respectively) and 95% confidence intervals (dotted black lines) based on the available points are superimposed. The projections intersect just before the 2156 Olympics, when the winning women's 100-metre sprint time of 8.079 s will be faster than the men's at 8.098 s.

Coefficient of Determination (R^2)

- The strength of a linear model is commonly evaluated using R^2.
- R^2 measures the **proportion of variability in** y **explained** by the model.
- The remaining variability is due to:
  - variables not included in the model, or
  - inherent randomness.
- In (only) simple linear regression:

$$R^2 = r^2$$

- Example:

$$R^2 = (-0.75)^2 = 0.56$$

Computing R^2

- R^2 can also be written using sums of squares:

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SST - SSE}{SST}$$

- $SST = \sum(y_i - \bar{y})^2$ (total variability)
- $SSE = \sum(y_i - \hat{y}_i)^2$ (unexplained variability)

- SST (Total Sum of Squares) - SSE (Sum of Squared Errors) is the variability **explained by the model**.

R^2

- The strength of the fit of a linear model is most commonly evaluated using R^2.
- R^2 is calculated as the square of the correlation coefficient.
- It tells us what percent of variability in the response variable is explained by the model.
- The remainder of the variability is explained by variables not included in the model or by inherent randomness in the data.
- For the model we've been working with, $R^2 = (-0.75)^2 = 0.56$.

Coefficient of Determination

- The coefficient of determination is denoted by R^2.
- It measures the proportion of variability in y explained by the fitted model.

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SST - SSE}{SST}$$

where

$$SSE = \sum (y_i - \hat{y}_i)^2$$

and

$$SST = \sum (y_i - \bar{y})^2$$

Interpretation of R^2

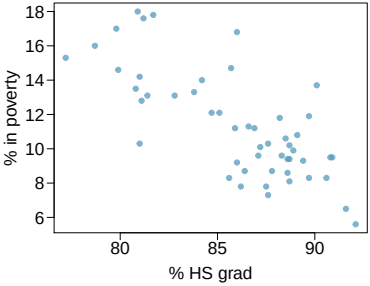
Which of the below is the correct interpretation of $r = -0.75$, $R^2 = 0.56$?

(a) 56% of the variability in the % of HG graduates among the 51 states is explained by the model.

(b) 56% of the variability in the % of residents living in poverty among the 51 states is explained by the model.

(c) 56% of the time % HS graduates predict % living in poverty correctly.

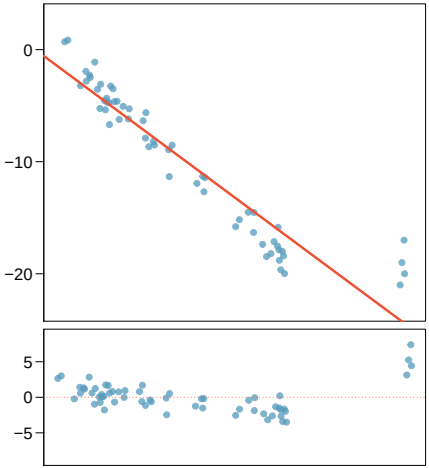
(d) 75% of the variability in the % of residents living in poverty among the 51 states is explained by the model.



Types of outliers

How do outliers influence the least squares line in this plot?

To answer this question think of where the regression line would be with and without the outlier(s).



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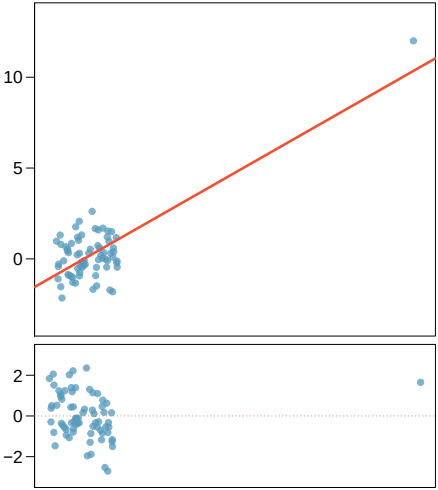
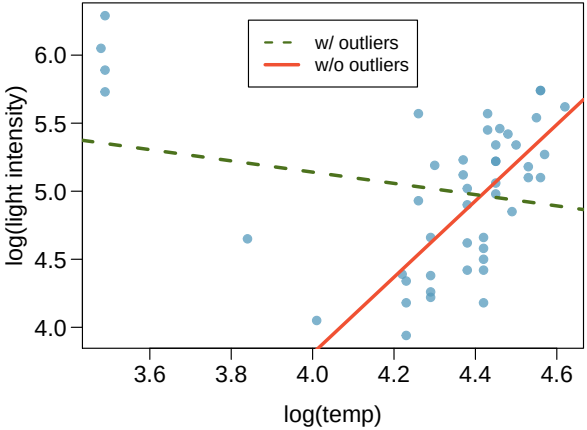


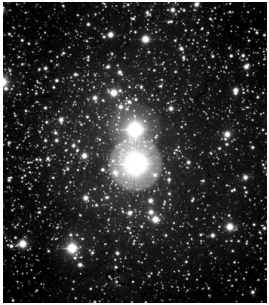
Some terminology

- **Outliers** are points that lie away from the cloud of points.
- Outliers that lie horizontally away from the center of the cloud are called **high leverage** points.
- High leverage points that actually influence the slope of the regression line are called **influential** points.
- In order to determine if a point is influential, visualize the regression line with and without the point. Does the slope of the line change considerably? If so, then the point is influential. If not, then it?s not an influential point.

Influential points

Data are available on the log of the surface temperature and the log of the light intensity of 47 stars in the star cluster CYG OB1.

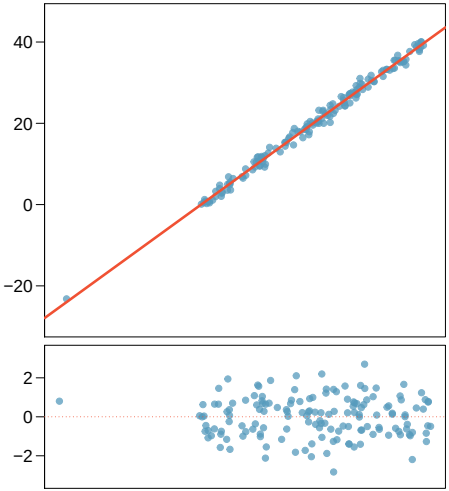




Types of outliers

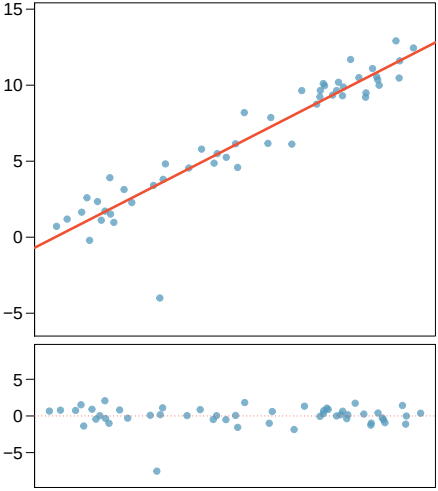
Which of the below best describes the outlier?

(a) influential
(b) high leverage
(c) none of the above
(d) there are no outliers



Types of outliers

Does this outlier influence the slope of the regression line?



Recap

Which of following is true?

(a) Influential points always change the intercept of the regression line.

(b) Influential points always reduce R^2.

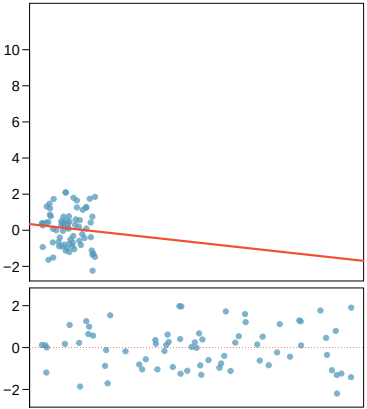
(c) It is much more likely for a low leverage point to be influential, than a high leverage point.

(d) When the data set includes an influential point, the relationship between the explanatory variable and the response variable is always nonlinear.

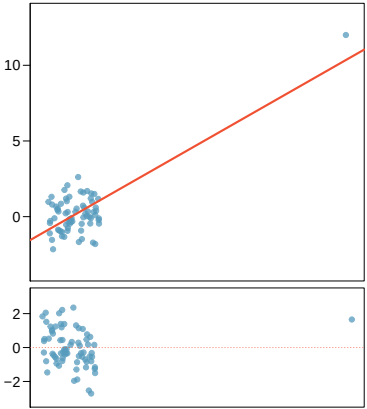
(e) None of the above.

Recap (cont.)

$R = 0.08, R^2 = 0.0064$



$R = 0.79, R^2 = 0.6241$



Error and Sum of Squares

- Residual: $e_i = y_i - \hat{y}_i$
- Error Sum of Squares:

$$SSE = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$$

Equivalent form:

$$SSE = S_{yy} - b_1 S_{xy}$$

Measures unexplained variability.

Estimating σ^2

- σ^2: true variance of errors (unknown)
- Estimate using:

$$MSE = s^2 = \hat{\sigma}^2 = \frac{SSE}{n-2}$$

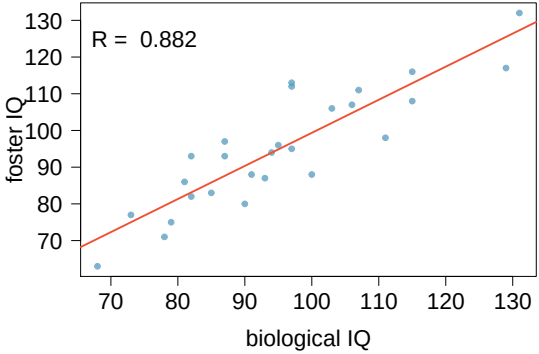
- Equivalently:

$$MSE = \frac{S_{yy} - b_1 S_{xy}}{n-2}$$

Why $n-2$?
Two parameters estimated: b_0, b_1

Nature or nurture?

In 1966 Cyril Burt published a paper called “The genetic determination of differences in intelligence: A study of monozygotic twins reared together and apart”. The data consist of IQ scores for [an assumed random sample of] 27 identical twins, one raised by foster parents, the other by the biological parents.



Which of the following is <u>false</u>?

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       9.20760    9.29990   0.990    0.332
bioIQ             0.90144    0.09633   9.358  1.2e-09

Residual standard error: 7.729 on 25 degrees of freedom
Multiple R-squared: 0.7779,Adjusted R-squared: 0.769
F-statistic: 87.56 on 1 and 25 DF,  p-value: 1.204e-09
```

(a) Additional 10 points in the biological twin's IQ is associated with additional 9 points in the foster twin's IQ, on average.

(b) Roughly 78% of the foster twins' IQs can be accurately predicted by the model.

(c) The linear model is $\widehat{fosterIQ} = 9.2 + 0.9 \times bioIQ$.

(d) Foster twins with IQs higher than average IQs tend to have biological twins with higher than average IQs as well.

Inference for the Slope

The standardized variable

$$T = \frac{B_1 - \beta_1}{s/\sqrt{S_{xx}}}$$

has a t-distribution with $n - 2$ degrees of freedom.

$$s = \sqrt{\frac{SSE}{n-2}} == \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-2}}$$

This result is the basis for:

- confidence intervals for β_1
- hypothesis tests for β_1

Confidence Interval for the Slope

A $100(1-\alpha)\%$ confidence interval for β_1 is

$$b_1 - t_{\alpha/2,n-2}\frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2,n-2}\frac{s}{\sqrt{S_{xx}}}$$

where $t_{\alpha/2,n-2}$ is the critical value from the t-distribution with $n-2$ degrees of freedom.

Compare to estimating μ (Chapter 9):

$$\bar{x} - t_{\alpha/2,\,\nu}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2,\,\nu}\frac{s}{\sqrt{n}},$$

where $\nu = n-1$

Hypothesis Test for the Slope

Null hypothesis:

$$H_0 : \beta_1 = \beta_{10}$$

Test statistic:

$$t = \frac{b_1 - \beta_{10}}{s/\sqrt{S_{xx}}} \sim t_{n-2}$$

Alternatives and rejection regions:

$$H_1 : \beta_1 > \beta_{10} \implies t \geq t_{\alpha,n-2}$$

$$H_1 : \beta_1 < \beta_{10} \implies t \leq -t_{\alpha,n-2}$$

$$H_1 : \beta_1 \neq \beta_{10} \implies |t| \geq t_{\alpha/2,n-2}$$

Significance Test of the Linear Model

- A common test is

$$H_0 : \beta_1 = 0 \qquad \text{vs} \qquad H_1 : \beta_1 \neq 0$$

- If we reject H_0, the slope is nonzero.
- Then the linear regression model is said to be **statistically significant**.

Inference for the Intercept

The standardized variable

$$T = \frac{B_0 - \beta_0}{s\sqrt{\frac{\sum x_i^2}{nS_{xx}}}}$$

has a t-distribution with $n - 2$ degrees of freedom.

Confidence Interval for the Intercept

A $100(1-\alpha)\%$ confidence interval for β_0 is

$$b_0 - t_{\alpha/2,n-2} s \sqrt{\frac{\sum x_i^2}{nS_{xx}}} < \beta_0 < b_0 + t_{\alpha/2,n-2} s \sqrt{\frac{\sum x_i^2}{nS_{xx}}}$$

Hypothesis Test for the Intercept

Null hypothesis:

$$H_0 : \beta_0 = \beta_{00}$$

Test statistic:

$$t = \frac{b_0 - \beta_{00}}{s\sqrt{\frac{\sum x_i^2}{nS_{xx}}}} \sim t_{n-2}$$

Alternatives:

$$H_1 : \beta_0 > \beta_{00}, \quad H_1 : \beta_0 < \beta_{00}, \quad H_1 : \beta_0 \neq \beta_{00}$$

- In practice, we are usually less interested in the intercept than the slope.

Prediction Interval for a Future Observation

The variable

$$T = \frac{\hat{Y}_0 - Y_0}{s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}$$

has a t-distribution with $n - 2$ degrees of freedom.

Therefore, a $100(1 - \alpha)\%$ prediction interval for Y_0 is

$$\hat{y}_0 \pm t_{\alpha/2, n-2} s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Prediction

- One purpose of regression is to predict response values at specified x-values.
- Two common inferential goals:
  - Confidence interval for the mean response $\mu_{Y|x_0}$
  - Prediction interval for a future single observation Y_0

Testing for the slope

Assuming that these 27 twins comprise a representative sample of all twins separated at birth, we would like to test if these data provide convincing evidence that the IQ of the biological twin is a significant predictor of IQ of the foster twin. What are the appropriate hypotheses?

(a) $H_0 : b_0 = 0$; $H_A : b_0 \neq 0$

(b) $H_0 : \beta_0 = 0$; $H_A : \beta_0 \neq 0$

(c) $H_0 : b_1 = 0$; $H_A : b_1 \neq 0$

(d) $H_0 : \beta_1 = 0$; $H_A : \beta_1 \neq 0$

Testing for the slope (cont.)

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	9.2076	9.2999	0.99	0.3316
bioIQ	0.9014	0.0963	9.36	0.0000

- We always use a t-test in inference for regression.

  Remember: Test statistic, $T = \frac{\text{point estimate} - \text{null value}}{SE}$

- Point estimate = b_1 is the observed slope.
- SE_{b_1} is the standard error associated with the slope.
- Degrees of freedom associated with the slope is $df = n - 2$, where n is the sample size.

  Remember: We lose 1 degree of freedom for each parameter we estimate, and in simple linear regression we estimate 2 parameters, β_0 *and* β_1.

Testing for the slope (cont.)

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	9.2076	9.2999	0.99	0.3316
bioIQ	**0.9014**	*0.0963*	**9.36**	0.0000

Confidence interval for the slope

Remember that a confidence interval is calculated as *point estimate* $\pm$ *ME* and the degrees of freedom associated with the slope in a simple linear regression is $n - 2$. Which of the below is the correct 95% confidence interval for the slope parameter? Note that the model is based on observations from 27 twins.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	9.2076	9.2999	0.99	0.3316
bioIQ	0.9014	0.0963	9.36	0.0000

(a) $9.2076 \pm 1.65 \times 9.2999$

(b) $0.9014 \pm 2.06 \times 0.0963$

(c) $0.9014 \pm 1.96 \times 0.0963$

(d) $9.2076 \pm 1.96 \times 0.0963$

Recap

- Inference for the slope for a single-predictor linear regression model:
  - Hypothesis test:

$$T = \frac{b_1 - null\ value}{SE_{b_1}} \qquad df = n - 2$$

  - Confidence interval:

$$b_1 \pm t^{\star}_{df=n-2} SE_{b_1}$$

- The null value is often 0 since we are usually checking for **any** relationship between the explanatory and the response variable.
- The regression output gives b_1, SE_{b_1}, and **two-tailed** p-value for the t-test for the slope where the null value is 0.
- We rarely do inference on the intercept, so we'll be focusing on the estimates and inference for the slope.

Caution

- Always be aware of the type of data you're working with: random sample, non-random sample, or population.
- Statistical inference, and the resulting p-values, are meaningless when you already have population data.
- If you have a sample that is non-random (biased), inference on the results will be unreliable.
- The ultimate goal is to have independent observations.